From policy documents to ODRL

Literature, software and a concrete implementation proposal

Chak Wong

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Abstract

The 2023 SFC–HKMA suitability circular requires more than extraction of permissions and prohibitions: its streamlined procedures depend on eligibility, prior consent, conditional relief and continuing duties. This review examines text-to-ODRL methods, regulatory profiles, formal semantics and available software, then proposes a bounded extension to LegalMath using its existing prospectus evidence machinery and SPI calculations. ODRL is the first proposed output of a reviewed source-linked interpretation; Catala supplies selected computations and additional checks. The proposal specifies the profile, modules, result contracts, worked export, implementation sequence and counterexamples needed to evaluate the feature. It distinguishes implemented repository components from proposed ODRL work and structural conformance from legal fidelity.

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1 From policy documents to ODRL: what must survive translation

The next proposed output of LegalMath is a machine-readable policy in the Open Digital Rights Language, ODRL. The immediate case is the joint SFC–HKMA circular of 28 July 2023, reference 23EC35, on the Streamlined Approach for sophisticated professional investors (SPIs). The useful question is whether an intermediary may apply particular streamlined procedures to a client and transaction, and which duties remain. It is not simply whether a trade is allowed. This difference determines what the translation must express (SFC and HKMA, 2023a,b,c).

Consider a designated account whose gross exposure has risen above the client's streamlining threshold. A sentence-level translator might read the threshold rule and produce a prohibition on every further transaction. Annex 2 FAQ 6 expressly permits the account to continue operating and executing transactions while top-ups and deposits are restricted. It does not expect the intermediary to reduce or unwind exposure for this purpose. An ordinary suitability route also remains relevant. The missing distinction is the kind of action and the monitoring arrangement, not a different spelling of the threshold (SFC and HKMA, 2023c, FAQ 6).

This review therefore recommends *ODRL as the first new deliverable*, produced from a reviewed interpretation linked to the source. Selected Catala calculations should accompany it. Requiring a complete Catala program first would postpone the desired policy representation without resolving its normative questions. Treating ODRL as the only representation of the circular would instead risk losing meanings that the selected ODRL profile cannot execute. The implementation proposal below develops a shared interpretation with explicit projections into these different languages.

1.1 Scope and strength of the evidence

The review was completed on 22 October 2026. It combines the circular body, both annexes, direct text-to-ODRL research, work on regulatory profiles and formal semantics, and static inspection of selected software. The official annex PDFs match the repository's archived copies byte for byte. This fixes a historical translation case; a current operational application would still need the relevant source editions, amendments and incorporated provisions. The Code of Conduct, Professional Investor Rules and referenced FAQs must remain named dependencies rather than silently assumed complete (LegalMath research record, 2026b).

Searches used Crossref, OpenAlex, GitHub, the W3C ODRL community's implementation landscape, and references from the inspected papers. The recorded "HKMA ODRL" query returned no results, and no direct HKMA translation study was found in the inspected material. This is a targeted survey, not proof that no such work exists. Eight newly retained papers received technical reading of the relevant method, semantics and evaluation; existing Catala and eFLINT papers supplied focused comparisons. Nine public code checkouts were pinned. Unavailable originals and documentation-

only findings remain identified in the accompanying reading record. No generation experiment or third-party software benchmark was run (LegalMath research record, 2026b).

The important evaluation question is whether a translation preserves the decision and duty distinctions in the circular. A graph can be valid RDF, conform to a set of shapes, and still prohibit the transactions that FAQ 6 allows. Conversely, a disagreement between evaluators can reflect different declared semantics rather than a malformed graph. The following discussion keeps these questions separate so that published scores are not asked to answer a question their experiments did not test.

1.2 ODRL's basic model and the need for a profile

ODRL is a W3C model for expressing permissions, prohibitions and duties concerning identifiable resources. An action says what may, may not, or must be done; a party identifies a participant; an asset identifies the resource to which the action applies. Constraints restrict when a rule applies. The same policy can be serialized as JSON-LD, a JSON representation that maps names to identifiers in an RDF graph. RDF supplies the linked statements; it does not infer the legal meaning of "sophisticated investor" (W3C, 2018, secs. 2–3).

The distinction between two ODRL relationships is particularly important. A duty attached to a permission through `odrl:duty` is a prerequisite for that permission. A duty connected directly to a policy through `odrl:obligation` is an independent obligation. Prior written consent and a later annual review should therefore not be represented as an undifferentiated list of actions that must all have happened before the first transaction. The circular and the chosen event interpretation must determine which performance is needed, by whom and at what time (W3C, 2018, secs. 2.6.1, 2.6.4–2.6.5).

A *profile* defines additional vocabulary and semantics for a particular application. The proposed HKMA profile would name applying the Streamlined Approach, reviewing continued qualification, providing a requested explanation, and the observations used to establish their conditions. ODRL requires a processor to understand a declared profile and stop when it does not recognise it. A new identifier in a JSON file is consequently not enough: the evaluator needs an implementation of that term's meaning. A profile extends the core; it is not permission to reverse the meaning of an existing ODRL relationship (W3C, 2018, sec. 3).

For this case, the resource can be the identified service or procedure instance to which streamlined treatment applies. The intermediary is the party exercising that permission; the client and account remain separately identified. Regulatory source authority should be recorded explicitly in provenance. It should not be inferred merely by putting the regulator into an `assigner` field, which would hide a substantive mapping choice. A general regulatory rule set can use `odrl:Set`; a particular agreement with a client is a different record. These are proposed domain choices to specify and test, rather than universal mappings supplied by ODRL.

1.3 The circular contains calculations, dispensations and legal changes

The financial limb illustrates the calculational part. Let P be the qualifying portfolio value and N the qualifying net assets excluding the primary residence, both expressed in Hong Kong dollars at the declared assessment date. After the ownership, aggregation and currency questions have been resolved, the financial condition is

$$F(P, N) = (P \geq 40{,}000{,}000) \ \vee \ (N \geq 80{,}000{,}000). \tag{1}$$

Equality satisfies either threshold. A portfolio of HK$40 million therefore passes this limb even when the net-asset alternative does not. The disjunction does not establish SPI status: professional-investor status, knowledge or experience and investment objectives are additional conditions. Joint holdings, written allocations, liabilities and qualifying investment-holding corporations require their own source-based treatment (SFC and HKMA, 2023b, paras. 1–5).

Knowledge or experience can be established through the specified degree or postgraduate diploma, a professional qualification in finance, at least one year of relevant professional work in the financial sector, or at least five transactions within the past three years in the same product category. Reliance on trading experience therefore needs category identity and a calendar reference date. A representative's expertise is not automatically the client's expertise, although transactions through a representative or joint account may contribute to the client's trading history. Unknown objectives cannot establish that the client is nonconservative (SFC and HKMA, 2023b, paras. 4.1, 5.1 and 7.3) (SFC and HKMA, 2023c, FAQ 2).

The procedure relaxations introduce a different kind of statement. Paragraphs 10 and 11 say that specified procedures are "not required" under the applicable conditions. This is a dispensation from a requirement; it is not a prohibition on performing that procedure. Nor is it a blanket exemption from suitability obligations. The explanation relaxation has exceptions for requests and material queries, and solicited or recommended transactions differ from unsolicited complex-product transactions. A translation must preserve the scope of the relieved duty and the duties that survive (SFC and HKMA, 2023b, paras. 10–11).

Client choices also change what applies. Prior written agreement and the specified explanation precede use of the approach. Paragraph 13.1(d) provides for withdrawal; paragraphs 13.2(c)(iii) and 13.2(d)(iii) concern changes to categories and thresholds; paragraph 14 requires annual review and reminders. An old consent event cannot authorize indefinitely after withdrawal. Equally, "annually" does not choose a 365-day arithmetic convention. The selected calendar and effective-time interpretation must be visible (SFC and HKMA, 2023b, paras. 13–14).

Finally, the financial evidence requirement must not be strengthened accidentally. FAQ 1 permits reliance on disclosed information after reasonable know-your-client efforts in the stated circumstances; inconsistencies need clarification. Recordkeeping and universally demanding documentary proof are different requirements. These

qualitative assessments need evidence and a declared interpretation. Inventing a numeric cutoff for "reasonably satisfied" would make a program easier to run by changing its subject (SFC and HKMA, 2023c, FAQ 1).

1.4 What direct text-to-ODRL research demonstrates

Revel et al. (2023) offer an inspectable translation route: a sentence is parsed into Abstract Meaning Representation (AMR), converted into RDF, and transduced into ODRL. AMR makes linguistic relationships explicit before the policy representation is selected. This is a useful baseline for separating linguistic analysis from policy construction. It does not remove the need to resolve definitions, references and the legal effect of exceptions.

Their experiment contains approximately 100 simple deontic sentences, about 20% handcrafted and the remainder generated from a grammar, plus ten negative examples. Modality extraction has reported precision 0.780, recall 0.640 and F-score 0.703; action extraction has precision 0.989, recall 0.810 and F-score 0.891. These are component scores rather than whole-policy legal accuracy. Document preprocessing and definition resolution are described as hypothetical extensions. The error discussion includes the scope of negation: permission not to reproduce is different from lack of permission to reproduce. That problem is directly relevant to the circular's "not required" language (Revel et al., 2023, method, experiments and error discussion).

Mustafa et al. (2024) take an ontology-guided language-model approach. They provide ODRL concepts, curated explanations and examples, then refine generated policies. Twelve cultural-dataspace use cases, three models and three methods give 108 configurations; the paper describes initially running each use case once. Its prominent 91.95% figure is 217 of 236 selected checks for one use case with refinement. It is not the probability that a complete legal policy was interpreted correctly. The method supports a small generation baseline, especially because it makes the allowed vocabulary explicit (Mustafa et al., 2024, secs. 2–4 and Table 1).

The associated LLM4ODRL code parses policies and checks SHACL shapes. SHACL expresses conditions on an RDF graph: for example, whether a particular kind of node has the required properties. Some repository shapes require creator, description, title and issue date. Those can be useful application requirements, but should not all be attributed to the W3C core standard. No repository-level licence was found in the inspected checkout, so the method can inform a baseline without assuming permission to copy its code (LegalMath research record, 2026a).

Zhong et al. (2026) add an orchestrator, reference rewriting, decomposition, ontology-guided generation and SHACL-guided correction. Seventy seed cases are expanded with a model into 770 cases: 440 simple, 220 with parallel relationships and 110 with sequential dependencies. The semantic evaluation uses checkpoints extracted by a model and judgments from two language models; the grammar score uses selected shape violations. These measurements are useful for the authors' generation experiment. They are not an independently established legal interpretation of a circular and its annexes.

AgentODRL also gives a reason to keep the first experiment simple. Fixed preprocessing workflows achieve higher aggregate semantic scores than the adaptive orchestrator in parts of the reported comparison. The evidence does not justify assuming that more orchestration improves this task. Compare a direct ontology-guided proposal with explicit reference handling before adopting the full system. Its Apache-licensed code is reusable, but the inspected correction loop can retain a final candidate after exhausting its correction budget without establishing success. The local result must preserve that failure status. Reference rewriting must also retain the passages that supplied each condition and exception (Zhong et al., 2026; LegalMath research record, 2026a).

There is earlier NLP work on RDF licence generation by Cabrio, Aprosio and Villata, discussed in the AMR paper. Its original technical text was not obtained in this survey, so it is a historical lead rather than separately verified performance evidence. Likewise, De Vos, Kirrane, Padget and Satoh's 2019 *ODRL Policy Modelling and Compliance Checking* is a directly relevant lead on regulatory profiles and answer-set-program compliance, including GDPR. The publication record and later references were located, but the original attachment remained unavailable. Detailed claims about its encoding or adoption would require that text (Revel et al., 2023; LegalMath research record, 2026b).

1.5 Legal meaning and formal semantics constrain the translation

Kebede et al. (2021) explain why a legal relationship can exceed a simple usage permission. Duties, institutional powers, delegation, revocation and policy lifecycle require distinctions that are easily hidden by a single allow/deny result. For this circular, withdrawing consent changes the arrangement; a dispensation changes which procedure is required; and the regulator, intermediary and client occupy different roles. The critique identifies modelling questions. Later semantics work must be considered alongside it rather than assuming ODRL has remained unchanged.

The inspected AIUP profile supplies a practical example of regulatory specialisation. It defines vocabulary for intended and precluded AI uses, oversight and reporting, while expressly limiting its coverage of the EU AI Act. Its draft specification, rather than the associated research paper, was the source inspected here. Its relevance is the discipline of declaring exact domain terms and supported questions. A corresponding HKMA profile should state which duties and changes it can represent, rather than call every operation `odrl:use` (Golpayegani et al., 2024).

Bonatti et al. (2025) develop declarative semantics for rule activation, action matching, duties and compliance over traces. A trace records relevant events and their order. This makes an otherwise hidden question explicit: does one performance of a duty permit one subsequent action or many? A consent to a continuing arrangement and an explanation in response to a particular request require different answers. Their discussion also distinguishes constraints on activation from conditions of fulfilment, considers policy gaps, and identifies unresolved features such as temporal operands and a separate meaning for `andSequence`. These are proposed semantic choices, not a new W3C Recommendation (Bonatti et al., 2025, sec. 3, especially 3.6–3.8).

Salas et al. (2026b) show how a restricted policy fragment can be normalised and compared under declared semantics. Their algorithms partition numerical and symbolic conditions into matching regions and establish preservation results. This suggests future checks for overlapping threshold conditions or an unintended change between exports. The fragment excludes attached duties, remedies and consequences; standalone obligations remain unchanged in the policy-preservation result. Splitting one obligation into several required obligations can change its meaning. The normal form can grow exponentially in the number of attributes, and empirical performance evaluation is left for future work. This is a basis for restricted comparison, not a generic equivalence procedure for the whole circular (Salas et al., 2026b, secs. 3–4, especially Theorem 6).

LegalRuleML and eFLINT help identify what the internal interpretation should retain even when the initial export is narrower. LegalRuleML addresses source correspondence, context, defeasibility, authority and alternative readings. The original eFLINT work separates action compliance from duty compliance and represents events that create or terminate normative relationships. Thus a transaction decision and an outstanding review duty can be reported separately. These ideas support the internal design; this proposal does not make a new LegalRuleML serializer or eFLINT runtime a prerequisite (OASIS LegalRuleML Technical Committee, 2021; van Binsbergen et al., 2020).

1.6 Evaluation software: what to reuse and what to test first

Salas et al. (2026a) formalise access decisions and offline and streaming monitoring, and present the OVAL implementation. Its algorithm processes events while tracking rule matches and outstanding requirements. That is closer to the circular's combination of procedure eligibility and continuing duties than syntax validation alone. The reported evaluation uses synthetic workloads and comparison with the SolidLab/FORCE cases. Agreement is 54 of 68 cases after mapping the comparison system's "not enough information" to false. Different reasoning and duty interpretations account for disagreements. Runtime results on these workloads therefore cannot establish greater legal correctness (Salas et al., 2026a, secs. 4–6).

The pinned October OVAL code is a reasonable first candidate for a bounded trial. Its access API defaults to accepting unfulfilled duties on a promise of future fulfilment; another setting requires prior performance. The default must not decide whether written consent can be obtained after the transaction. Missing values can become false or remain null depending on configuration; `andSequence` is implemented as conjunction rather than temporal sequence; event counting is not a category-specific rolling three-year history. Numeric comparisons attempt floating-point conversion, even where older feature documentation describes lexical comparison. Explicit units, some temporal operands and other constructs are unsupported. The proposal therefore supplies exact calculations and a declared event adapter, with unsupported terms reported before evaluation (LegalMath research record, 2026a).

FORCE offers a useful testing pattern: a policy, request, state of the world and expected compliance report. Its current documentation describes limitations on duties, while code includes duty-reporting structures and rules. Reporting a duty and enforcing it

as a prerequisite to permission are distinct capabilities. A helper also connects policy-level duties through `odrl:duty`, where the W3C model uses `odrl:obligation`. These are concrete compatibility checks for the intended input route, not evidence that every FORCE evaluation fails. The reported 67/68 activation-similarity result is not a local result or a measure of fulfilled legal duties. The associated 2025 papers were located bibliographically but not obtained in full, so the recommendation rests on inspected code and documentation (LegalMath research record, 2026a).

ODRE addresses the connection between descriptive policy terms and executable values. Cimmino et al. (2024) use runtime interpolation and operand/action adapters, with Python and Java implementations and 24 demonstration tests per implementation. Their tested enforcement is synchronous; temporal monitoring is future work in that version. The Python checkout presents additional static concerns: the rule traversal discards whether a rule was reached through permission, prohibition or obligation, and the constraint query appears to look directly on a rule where its own sample uses a nested constraint node. An empty constraint result is then accepted. These findings were not executed as failures, but they are sufficient to withhold selection as the first general normative evaluator pending focused reproduction (Cimmino et al., 2024; LegalMath research record, 2026a).

Eclipse Dataspace Components provides registered evaluation functions, execution scopes and JSON-LD policy transformation. These are useful later if the product needs dataspace negotiation or transfer services. They do not extract a circular's norms or define SPI qualification. pySHACL is the smaller immediate dependency for graph validation; its shapes must be reviewed and must actually target the intended policy nodes. The public ODRL-shape repository is a source of ideas, but no repository-level licence was found in the inspected checkout (LegalMath research record, 2026a).

Table 1: Selection for a first trial, based on static inspection rather than a benchmark ranking.

Component	Proposed use	Condition before reliance
AgentODRL	Generation and reference-handling ideas	Preserve source maps and failed correction status.
LLM4ODRL	Simple ontology-guided methodological baseline	Resolve licence before copying code; review shape assumptions.
pySHACL	Structural and profile checks	Review shapes, targeting and negative cases.
OVAL	First evaluator trial	Pin duty/default semantics; supply exact facts and explicit time handling.
FORCE	Scenario/report design and second evaluator	Establish duty and policy-obligation behavior on the chosen route.
ODRE	Operand-adapter design reference	Reproduce and resolve the inspected traversal concerns.
EDC	Later deployment integration	Supply the domain semantics; justify the larger integration cost.

The inspected AgentODRL, pySHACL documentation, ODRE and EDC declare Apache-2.0 licensing; OVAL and FORCE use MIT. The table links to the public repositories. The proposed first OVAL trial is pinned to `b2838782b7ce`; the maintained

AgentODRL checkout is `d55ed231ba7d`, FORCE is `194894a04588`, and ODRE Python is `1a2cb06004f0`. All nine full checkout commits and inspected-file hashes are retained in the survey manifest. These snapshots fix what was examined, not a promise about a later release. No software is ranked as production ready on this evidence (LegalMath research record, 2026a).

1.7 Choosing the relationship with Catala

Catala is valuable where the circular gives computations with defaults and exceptions. Equation 1, explicit allocation rules and category histories are candidates once their premises are fixed. The Catala paper's compilation result concerns translation between formal calculi. It does not prove that a natural-language source was interpreted correctly or that another language has the same semantics (Merigoux et al., 2021, secs. 4.2–4.5).

Translating the document into Catala first gives early executable calculations and uses existing tooling. It is attractive for a predominantly computational source or an already reviewed Catala model. Here, however, a calculation need not retain the distinction between a dispensation, a legal power to withdraw, and an outstanding duty. Reconstructing that information later for ODRL can lose meaning. Catala-first is therefore an unnecessary prerequisite for this circular.

ODRL-first followed by Catala delivers the desired interchange representation early and makes parties, actions and duties visible. But any meaning omitted from ODRL is also missing from the subsequent calculation. Agreement between the two generated outputs mainly tests that translation step. Fully separate document-to-ODRL and document-to-Catala processes provide more independence, but duplicate interpretation work and permit sources, terminology and assumptions to drift. They are useful experimental comparators on a small fragment, rather than the preferred everyday authoring process.

The recommended design retains a richer reviewed interpretation and exports ODRL while invoking Catala for selected calculations. Independent scenario specifications challenge both. This separates four questions: whether the graph conforms to its profile, whether the evaluator implements the chosen semantics, whether the two computations agree on a declared domain, and whether the interpretation is supported by the circular. Catala can help answer the third and expose defects relevant to the second. It does not establish the first or fourth by itself.

The practical consequence is to begin with source coverage and the FAQ 6, consent and explanation counterexamples. The initial feature should produce a reviewable policy and an explicit account of what it cannot express. A generic ODRL-to-Catala compiler is not required to obtain that result. Section 2 specifies the records, modules and checks needed to make this recommendation concrete.

2 Proposed feature: source-linked ODRL for the SPI circular

The proposed feature accepts a versioned policy-document bundle and produces a draft ODRL policy together with the source interpretation, supported scope and evaluation evidence needed to inspect it. The first bundle is circular 23EC35 and its two annexes. The first question is whether the Streamlined Approach is available under stated premises and what particular duties remain. The feature should not label its answer as general permission to trade. It should retain unresolved interpretations and observations as part of the answer rather than replace them with plausible defaults.

The review above supplies the basis for this design. Ontology-guided drafting is an established research approach, but its reported structural and model-judged scores do not establish source fidelity. ODRL evaluators also make different choices about duties and missing information. The proposed extension therefore adds deterministic policy export and explicit evaluator adapters around an interpretation that remains richer than the executable fragment. All ODRL modules, commands and examples in this section are proposed work. The source, translation and SPI components identified as existing were inspected in the repository; no ODRL runtime was executed for this proposal (LegalMath research record, 2026b,a).

2.1 Start from the work that already exists

The inspected main commit is `3c3e462acabf`; the prospectus branch `feature/prospectus-evidence-master` is at `e8beb52b1339`. That prospectus commit is an ancestor of the inspected main commit, and the committed `src/legalmath/prospectus` and `src/legalmath/translation` trees have no differences between them. The relevant code is already available on main. This comparison does not assert that unrelated worktree changes or every branch file are identical (LegalMath research record, 2026a).

The prospectus successor modules already preserve source identity and occurrences. `source_graph.py` retains document hashes, page and line locations, geometry and time information; `construction_ast.py`, `reference_closure.py` and `priorities.py` preserve source assembly, dependencies and proposed overrides. These mechanisms can keep an annex qualification attached to the paragraph it explains. They need new circular, FAQ and incorporated-source roles. A later date or a generic priority edge must not decide legal priority without an interpretation.

The clause graph provides a useful starting record for actors, affected parties, modality, conditions, exceptions and source spans. Its recogniser is currently a bounded bond grammar with effects such as principal loss, conversion and repayment. It is not already a general regulatory parser. The extension should reuse its source-occurrence checks while introducing regulatory norm types separately. This avoids turning a bond-specific `effect` field into an unexplained catch-all.

There is more direct reuse in `prospectus/models.py` and `prospectus/semantics.py`.

The financial model uses inclusive alternatives and integer Hong Kong dollar cents. The client model combines professional investor status, financial qualification, knowledge or experience and objectives. The threshold model already distinguishes a designated account without new funds from the ordinary exposure comparison. Selected duty models distinguish suitability matching, due diligence and explanation relief. These models explicitly identify themselves as conditional synthetic specifications. Their source bindings leave legal entailment and completeness unestablished; ownership, currency conversion, valid monitoring arrangements and other applicable requirements remain premises (LegalMath research record, 2026a).

Their exact threshold expression is particularly useful for a first regression case. Let G be gross exposure and M the streamlining threshold in integer HKD cents. Let D mean that the designated-account arrangement and its required controls are valid, and A mean that the request adds funds. Under those declared premises the existing amount-condition model is

$$T(G, M, D, A) = (G \leq M) \ \vee \ (D \wedge \neg A). \tag{2}$$

If G is HK$12 million and M HK$10 million, the first term is false. A transaction within a valid designated account without new funds can still satisfy the second term. A top-up does not. This is an illustrative conditional calculation: gross exposure must include leverage and the required treatment of transferred-out noncash investments, and D must not be set true merely because an account has a convenient name. The source for the exception must include FAQ 6, alongside the threshold and monitoring provisions (SFC and HKMA, 2023b, paras. 8.1–8.5) (SFC and HKMA, 2023c, FAQs 6–7).

The shared *LegalRuleModel* already retains typed facts, expressions, scope, source spans and interpretations. Its capabilities include exact amounts and declared date/collection operations on supported targets. `translation/pipeline.py` currently accepts only RuleIR and Catala; its build path assumes those executable targets. An ODRL export must therefore be a separate adapter with its own result type, not a new string added to the target list while retaining assumptions about Java builds. `translation/verification.py` accepts external expected outcomes and checks execution and replay; `prospectus/checks.py` contains independently written symbolic expressions and deliberate mutations. These are useful patterns, not an existing ODRL conformance suite (LegalMath research record, 2026a).

The lifecycle, duty-performance and amendment modules provide additional patterns for preserving events, evidence and changed decisions. They require explicit bindings to this circular. Prospectus analysis also contributes substantive evidence: Annex 2 FAQ 3 calls for distinct product categories for debt instruments with loss-absorption features. Evidence of contractual write-down or conversion can inform that classification. A paragraph match alone is insufficient; the operative product terms and reviewed category rule must support the assigned category (SFC and HKMA, 2023c, FAQ 3).

2.2 Keep norms and calculations connected but distinct

The proposed internal addition is a *PolicyInterpretation*, containing versioned norm records and links to existing calculation models. Each norm has a stable identity, exact source occurrences, the wording and interpreted modality, actor and affected parties, action and object, applicability, conditions, exceptions, timing and evidence requirements. It also records unresolved questions and the source of any priority. A duty records its trigger, required performance and how evidence establishes that performance. A client power records which state it changes, such as consent withdrawal. A dispensation names the particular duty it qualifies.

These records should reference typed calculations by model and rule identity. The financial norm can refer to the existing inclusive threshold calculation, while the interpretation retains the legal conditions for counting each holding. A generated Boolean called `eligible` is not enough: the record must preserve the premises that made it true and the source of each premise. Conversely, copying the complete computational expression into every ODRL constraint would obscure which part is a calculation and which is a normative consequence.

Every source unit must receive a disposition: represented in the supported export, retained as a definition or dependency, deliberately deferred with a reason, or unresolved. This makes the coverage denominator the source bundle, not the number of rules the generator happened to produce. The source graph must retain references to material outside the bundle. Missing dependencies can leave a conditional policy useful for research while preventing a claim to have represented every applicable requirement.

The following architecture keeps these responsibilities visible. The independent cases enter after the interpretation has been fixed; they must not be generated merely by reading back the same expression being tested.

2.3 Define two supported profiles before generating policies

The first implementation should distinguish an *eligibility profile* from a later *duty-monitoring profile*. These are proposed LegalMath profiles for the historical HKMA case, not profiles issued or endorsed by HKMA. Their versioned identifiers must bind vocabulary, shapes and executable semantics together. An evaluator must refuse an unknown profile or an unsupported construct before returning a legal-facing result.

The eligibility profile supports a permission to apply streamlined procedures to an identified request. It uses explicit intermediary, client, account, product-category and decision-time identities. Its conditions consume typed results for qualification, selected category, threshold treatment, continuing consent and other declared prerequisites. Monetary and calendar calculations are evaluated exactly outside the generic ODRL engine, with provenance and missing-fact status retained. This is an intentional computational boundary, not a claim that the engine itself understands joint ownership or currency conversion.

For the initial profile, use simple named constraints and explicitly defined conjunctions.

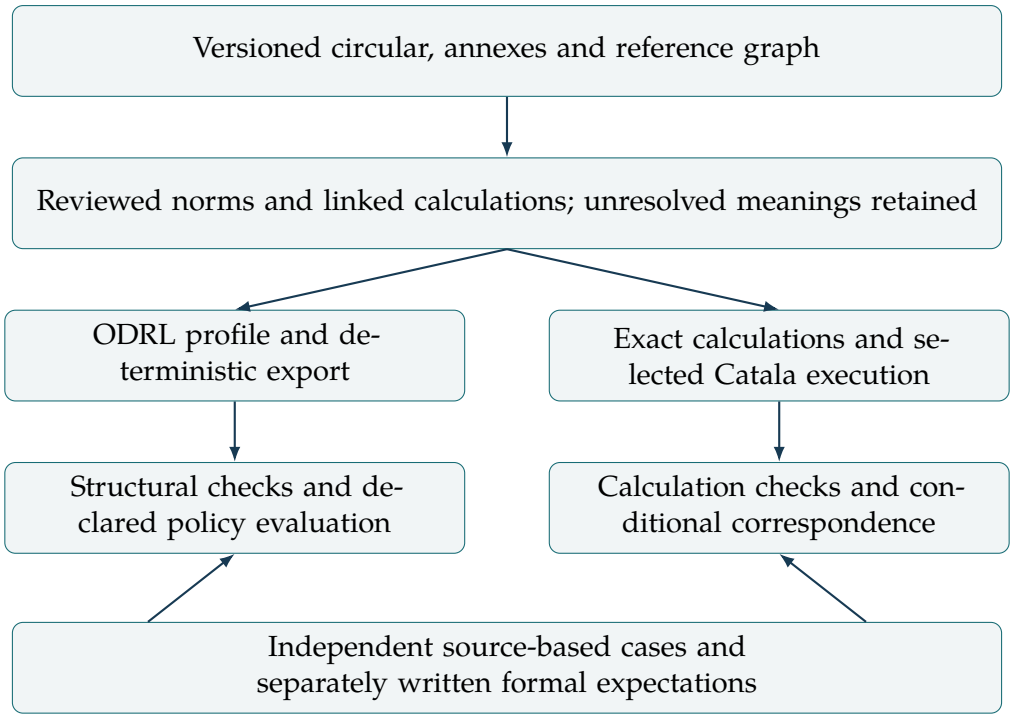


Figure 1: One source interpretation supplies complementary representations. Independent cases can expose a shared error; agreement between the outputs alone cannot.

Do not emit unsupported arithmetic, `inheritFrom`, `rightOperandReference`, temporal `andSequence`, remedies or consequences and hope that a consumer ignores them safely. An unknown term produces an unsupported result. A prohibition is exported only where a reviewed interpretation establishes the prohibited action and its scope; failure to establish streamlined eligibility does not create a trading prohibition.

The duty-monitoring profile extends this with named triggers and correlated performance events. Prior consent is verified as an effective prerequisite; annual review is a separately triggered obligation. An explanation duty is linked to the relevant request or material query and its required response. Evidence of a different explanation for a different transaction cannot satisfy it. Recurrence, deadlines and continued validity need explicit rules in the domain event adapter. A duty is fulfilled only when its required performance is established, rather than when the duty becomes active.

Dispensations require special care. The first implementation should calculate the applicable set of ordinary procedures and their conditional relief in the internal norm model, then export the supported remaining duties. It must retain a mapping from each relieved duty to the source and condition granting relief. If the ordinary duty's source has not been resolved, the result stays conditional. A custom *dispensation* extension can be considered later if interchange requires it, but only with defined semantics and evaluator support. It must never be approximated by a prohibition on the relieved action.

For general policies use `odrl:Set`, with regulatory authority in source provenance and party roles explicitly mapped by the profile. For a concrete export, the assignee is the intermediary and the target is the identified streamlined-service request. Client and

account identities remain bound in that request and its evidence record. The permission allows applying the procedure, so the target cannot silently change into the investment itself. These choices make the scope of the result inspectable.

2.4 A worked export makes the interface concrete

Suppose that a synthetic client and transaction satisfy the reviewed non-amount premises: professional-investor status, knowledge or experience, objectives, selected category and continuing prior agreement. The request is an internal transaction in a valid designated account, adding no funds, with the HK$12 million gross exposure and HK$10 million threshold described above. Equation 2 supplies the amount condition. The example also assumes the declared other requirements have been established; they are not automatically set true by the exporter.

The following JSON-LD illustrates the proposed eligibility-profile output. The `example.org` identifiers are reserved illustrative names. The inline context maps every abbreviated name to its RDF identifier, so the example does not rely on downloading an unspecified context. It is a design fragment, not a completed or conformance-tested translation of the circular.

Listing 1: An illustrative permission to apply a procedure to one request.

```
{
  "@context": {
    "odrl": "http://www.w3.org/ns/odrl/2/",
    "lm": "https://example.org/legalmath/",
    "prov": "http://www.w3.org/ns/prov#"
  },
  "@id": "lm:policy/spi-example-001",
  "@type": "odrl:Set",
  "odrl:profile": {
    "@id": "lm:profile/hkma-spi-eligibility-v1"
  },
  "prov:wasDerivedFrom": {
    "@id": "lm:interpretation/spi-001"
  },
  "odrl:permission": [{
    "@id": "lm:rule/apply-streamlined-001",
    "odrl:assignee": {"@id": "lm:party/intermediary-001"},
    "odrl:target": {"@id": "lm:request/transaction-001"},
    "odrl:action": {"@id": "lm:applyStreamlinedApproach"},
    "odrl:constraint": [{
      "odrl:leftOperand": {"@id": "lm:streamlinedEligible"},
      "odrl:operator": {"@id": "odrl:eq"},
      "odrl:rightOperand": true
    }]
  }]
}
```

The custom left operand is deliberately explicit. Its value is a query against the linked calculation and evidence record for this exact request, actor, client, account and decision time. The implementation must retain the component results and source locations behind it. A model-generated assertion that `streamlinedEligible=true` is not an acceptable substitute for that computation. The export manifest binds the policy bytes, profile version, norm model, calculation model and provenance map. A consumer without the profile implementation has a descriptive graph, not an executable permission.

On the declared facts the permission can apply. Changing only the request to a top-up makes the above-threshold designated-account alternative false. Changing only the evidence so that consent is missing makes the conclusion unresolved. Changing only the observation to a documented withdrawal makes continuing consent false after its effective time. None of these results, by itself, says that every ordinary-route trade is prohibited.

The result record should expose those distinctions through separate fields. The proposed permission result has one of `PERMITTED`, `NOT_ESTABLISHED`, `UNKNOWN`, `CONFLICT` or `UNSUPPORTED`, with rule and premise explanations. An explicit `PROHIBITED` result requires an applicable prohibition, not just the absence of a permission. `NOT_ESTABLISHED` here means that the declared streamlined route fails on known facts. It is not a claim about the ordinary route. Duty results are reported separately as inactive, pending, fulfilled, violated or unresolved under the declared event semantics. These result names are a proposed application contract, not new W3C vocabulary.

2.5 Consent, missing information and time need executable rules

The observation boundary should reuse the repository's distinction between known, unknown and conflicting evidence. The profile evaluates only supported observations whose subject, effective interval and knowledge time match the request. A known absence and an unobserved fact are different inputs. A complete history with four qualifying trades differs from an incomplete history that happens to contain four. A missing portfolio need not block the financial limb if the alternative net-assets branch is established, subject to the declared conflict policy. The existing complete and partial model profiles must not be mixed implicitly.

Consent requires an event model rather than an everlasting Boolean. Each record identifies the client, intermediary, agreed categories and threshold, source evidence, time of execution and effective interval. Withdrawal ends the relevant consent state according to the declared interpretation. Category or threshold changes produce new effective states while preserving earlier decisions. Replaying a Monday transaction should use what applied and was known on Monday; a later correction can produce a separate reassessment. The source supplies the rights and procedures, while exact timestamp and calendar conventions remain explicit implementation choices (SFC and HKMA, 2023b, paras. 13.1–14.2).

Annual review should create a due obligation according to a documented calendar rule and the relevant arrangement date. Its performance record must include the required

qualification/agreement checks and written reminders. The first release should require an explicit assessment of the due date where the source does not determine a unique calendar convention. Later calendar automation can replace that input after the convention is specified and checked. It must not make every future review a prerequisite to the first transaction or treat the first review as discharging every later one.

This is also why the OVAL trial must explicitly configure duty and unmatched-request behavior. Where a permission has an attached prerequisite, the adapter should request evidence of prior fulfilment and test that a future promise fails. The initial consent implementation can instead use a verified current-state condition, avoiding an assumption that one generic duty match models continuing consent. Either mapping needs the same withdrawal cases. The adapter must document which parts are evaluated by OVAL and which by the domain state processor; it must not describe the combined result as native support for a feature the engine lacks (LegalMath research record, 2026a).

2.6 Concrete modules and output contract

Implement the extension under a new `src/legalmath/odrl/` package. Keep the initial export API separate from the existing Java/Catala build function, reusing its model validation, source identities and hashing where appropriate. The following module boundaries are proposed to make the first change reviewable without refactoring unrelated execution code.

Table 2: Proposed modules and their first responsibilities.

Module	Responsibility and principal output
`model.py`	Versioned PolicyInterpretation and Norm records; source/exception links; typed references to existing calculation rules; coverage and unresolved questions.
`profile.py`	Supported vocabulary, norm forms, operators, party/target mapping, result semantics and versioned capability report.
`export.py`	Deterministic JSON-LD/Turtle export of the supported projection; source map and explicit unsupported/deferred dispositions.
`validation.py`	RDF parsing, core/profile shape checks, target-coverage checks and negative-case reports.
`facts.py`	Exact calculation results, currency and category identity, missing/conflicting observations, request bindings and valid/known times.
`events.py`	Consent and withdrawal state, category/threshold changes, review triggers and performance-event correlation.
`evaluators/oval.py`	Pinned evaluator interface with explicit duty/default settings, capability checks and normalized explanations.

Module	Responsibility and principal output
`verification.py`	Separately specified expected outcomes, fragment comparison, mutations and evidence of unresolved source questions.

Profile assets should live under `schemas/odrl/hkma-spi/`: a versioned context, vocabulary, reviewed SHACL shapes and capability declaration. Cases should live under `tests/fixtures/odrl/hkma-spi/`, with focused tests in `tests/odrl/`. These paths are proposed additions. The first test suite should exercise substantive distinctions and actual adapter boundaries, rather than mirror every helper function.

A proposed export directory contains `policy.jsonld`, optionally `policy.ttl`, `source-map.json`, `coverage.json`, `capabilities.json`, `validation.json` and `manifest.json`. Evaluation adds the observation/event snapshot and `evaluation.json`. The manifest records source and interpretation hashes, profile and shape versions, exporter identity, evaluator commit and settings, calculation model and build identity where used, and decision/knowledge times. Canonical JSON hashes identify exact serialized bytes. RDF graph equivalence is a different check, because harmless statement order or blank-node labels can change serialization; do not call a byte hash a semantic equivalence proof.

Expose three distinct operations: `export`, `validate` and `evaluate`. Export reports which norms were represented and which were not. Validation reports structural/profile findings. Evaluation reports permission and duty results with their premises. An exploratory draft may retain unsupported clauses, but a successful executable-profile result must not silently drop them. The first implementation should use a small Python API and local command wrapper; an HTTP endpoint can be added after the contracts stabilize. Proposed commands and result names must not appear in the current product documentation as already available.

2.7 What would count as evidence that the feature works

Four checks answer different questions. Parsing and reviewed SHACL shapes establish graph conformance. Tests against explicitly declared semantics check the evaluator and domain adapters. Independent implementations or bounded proofs compare computations. Source-linked analysis examines whether the formal interpretation is justified and complete within its declared scope. Passing one does not establish the next. Reviewer decisions and source-based case labels can challenge an interpretation, but cannot by themselves turn natural-language fidelity into a proved result.

For an executable correspondence claim, first fix the supported fragment, input domains, time convention and treatment of missing/conflicting evidence. Let N be the fixed norm model, $E(N)$ its ODRL export, and x a supported observation/event snapshot. Let $R(N, x)$ be a separately written reference specification, $V(E(N), x)$ the declared ODRL evaluation, and π the explicit projection to the outcomes both represent.

The intended engineering claim is

$$\forall x \in \mathcal{D}_{\text{supported}}, \qquad \pi\big(V(E(N), x)\big) = R(N, x). \tag{3}$$

Here π must preserve relevant result categories; it cannot erase unknowns or duty performance merely to obtain agreement. Finite tests establish this equality only for the cases run. Exhaustive finite-domain checking or a symbolic proof can establish it under the stated domain and encoding. Neither proves that N is the legally correct interpretation of the circular. For a Catala comparison, restrict both sides to the particular calculations and result semantics Catala implements. There is no proposed generic theorem equating all ODRL duties with all Catala programs.

Begin with the following source-based scenario families. Each needs an exact expected result, the relevant source passages, explicit premises and the reason that changing the fact changes the result. Where interpretation is open, retain alternative conditional expectations rather than invent one unqualified answer.

1. Financial boundaries: exactly HK$40 million, just below it, the HK$80 million alternative, both below, and missing data. Test inclusive comparison and disjunction separately from complete SPI qualification.
2. Qualifying facts: conservative and unknown objectives, missing professional-investor status, each knowledge/experience route, and category history with four versus five trades, wrong categories and calendar boundaries.
3. Ownership and product evidence: joint-account allocations with and without written allocation, primary-residence exclusion, and loss-absorption features incorrectly grouped with ordinary debt.
4. Consent and changes: missing prior consent, consent obtained later, continuing agreement, withdrawal and effective category/threshold amendments.
5. Procedure relief: requested explanation or material query, solicited versus unsolicited complex-product cases, and performance evidence for the particular surviving duty.
6. Threshold monitoring: execution-based versus designated-account arrangements, continued above-threshold transactions versus top-ups, leverage, transferred-out noncash exposure, and no invented unwind duty.
7. Continuing controls: review due versus completed, required reminders, repeated reviews, event order and replay with historical knowledge.
8. Evidence quality: reasonable KYC with consistent disclosures versus contradictory information requiring clarification, without an invented universal documentary-evidence requirement.

Use deliberate mutations to show that these cases detect the intended errors: replace $\geq$ with $>$; replace the financial "or" with "and"; omit consent; ignore withdrawal; drop FAQ 6; count transactions across unrelated categories; turn dispensation into prohibition; or convert unknown to false. The existing symbolic checks can supply a starting

pattern for the arithmetic fragment. Each mutation must be killed by a case whose expected behavior was specified independently of the mutated exporter.

An inverted modality, missing exception, fabricated prerequisite, silent unsupported construct or collapsed unknown blocks acceptance of the affected fragment. Structural success, a round trip, a model similarity score or speed cannot compensate. Report permission to use the approach, active duties, fulfilled or violated duties, and unresolved questions separately. Source coverage should likewise state represented, deferred and unresolved units; "100% of emitted rules validated" has no useful coverage denominator.

2.8 Implementation order and bounded recovery

The first milestone is a source inventory and manually specified norm model for the selected circular and annexes. Attach FAQ 6 to the amount rule and separate eligibility, dispensation and continuing duties. Reconcile each reused SPI fixture with its actual premises, adding missing source bindings without relabelling the synthetic fixture as legally certified. Completion means that every source unit has a disposition and every proposed executable norm has a supported or explicitly conditional interpretation.

The second milestone is the eligibility profile, deterministic exporter and structural checks. Its smallest useful end-to-end case is the above-threshold designated-account transaction and matching top-up, with exact financial preprocessing and missing-consent handling. This can be completed without any language-model generation. It isolates defects in the representation and evaluator interface before generation quality becomes a confounder.

The third milestone is a bounded OVAL trial, with pinned settings and the independent cases above. A mismatch must first be classified as an exporter bug, unsupported feature, adapter error or semantic disagreement. If an engine cannot support the declared fragment, retain the descriptive export and explicit unsupported result; narrow the executable fragment or implement the smallest specified adapter. Do not repeatedly patch examples until they agree while leaving the semantics undefined. FORCE can provide a second comparison once the particular duty and obligation behavior has been tested. An evaluator failure does not block independent work on source coverage or export validation.

The fourth milestone adds consent/withdrawal state and the duty-monitoring profile. Requested explanations and annual review should be separate event cases. Existing lifecycle and duty-performance patterns can be adapted here. This milestone depends on the norm model and event semantics, not on a successful generative model. It must demonstrate activation and performance separately and preserve a violation or unresolved obligation after the action has occurred.

Only after the manual path works should the fifth milestone compare generated proposals. Use the same source bundle and profile for a direct ontology-guided baseline and a reference-aware method inspired by AgentODRL and the source graph. Keep the evaluation cases out of generation prompts. Record model and prompt versions,

stochastic settings, budgets and all failed outputs. Predeclare a repeated-run protocol and uncertainty analysis before ranking methods. Published synthetic scores cannot justify a local default. A small trial can reject a structurally or semantically defective candidate without establishing that the surviving method is generally better.

The final milestone packages an explicit release candidate: source and interpretation versions, supported profile, deterministic export, evaluator configuration, result explanations, retained scenarios and known gaps. Where Catala adds an independent calculation check, record its exact scope. No successful milestone resets an unresolved source question or expands the claim to all policy documents. Generic regulatory profiles, a general ODRL-to-Catala compiler, dataspace deployment and automatic legal-priority resolution remain later choices requiring their own evidence.

The immediate implementation decision is therefore concrete: build the manual, source-linked eligibility export and the FAQ 6/consent cases first, using the existing SPI and source machinery. That produces the requested ODRL representation and exposes the semantic integration work at low cost. The duty extension and generation comparison follow on independent, declared prerequisites. The resulting feature can explain what a policy means under its premises, what the software checked, and what the evidence still leaves open.

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